# Task 10: RAG-Based Text Retrieval System

## Objective

To build a basic Retrieval-Augmented Generation (RAG) system that retrieves relevant information from a knowledge base using embeddings and similarity search.

## Technologies & Libraries Used

- Python
- sentence-transformers
- scikit-learn (cosine similarity)

## System Workflow

1. User inputs query
2. Query is converted into embeddings
3. Knowledge base is split into chunks (size = 600)
4. Chunks are converted into embeddings
5. Cosine similarity is computed
6. Best matching chunk is selected
7. Chunk is returned as answer

## Key Features Implemented

- Chunk-based retrieval system
- Embedding-based similarity search
- Dynamic answer generation (no hardcoding)
- Efficient and scalable pipeline

## Chunking Strategy

- Chunk Size: 600 characters
- Helps improve accuracy and performance
- Ensures better context matching

## Test Cases Implemented

- Valid query retrieval
- Different question formats
- Irrelevant queries (returns closest match)

## Errors Faced

- Incorrect chunk splitting
- Low similarity accuracy
- Embedding mismatch issues

## Solutions Implemented

- Fixed chunk size handling
- Improved similarity calculation using cosine similarity
- Optimized embedding generation

## Output

```
Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
(venv) PS D:\Daily Task\Task_10> python task10.py

🤖 NexusVortex Bot (Task 10)
Type 'exit' to quit

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 55/55 [00:00<00:00, 6093.15it/s]
BertModel LOAD REPORT from: sentence-transformers/paraphrase-MiniLM-L3-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
💬 Ask: I'm seeing Error XV-505 in my terminal. What is the root cause and how do I fix it?

📌 Answer:
 XV-505

⭐ Groundedness Score: 1/5
```

```
💬 Ask: My system is extremely laggy. Which shards should I be worried about and what is the specific command to reset them?

📌 Answer:
 1.2 Shard-Specific Responsibilities Each of the 12 shards is hard-coded to manage specific system silos to prevent resource contention:

⭐ Groundedness Score: 4/5

💬 Ask: Can I bypass the Theta-Sync requirement to install this on my old i7 laptop?

📌 Answer:
 Theta-Sync Mismatch: The installer detected a non-Vortex processor (x86/ARM64) lacking the required hardware-level instruction set.

⭐ Groundedness Score: 3/5

💬 Ask: I have a MAC address that was black-holed by the Wraith Firewall. Can I unblock it using the Galactic-Relay?

📌 Answer:
 ? The MAC Black-Hole: If the Wraith Firewall detects three unauthorized packets from a single MAC address within a 600-second window, it "Black-Holes" that add
ress.

⭐ Groundedness Score: 5/5
```

```
💬 Ask: How does the OS handle user login now that passwords and 2FA are deprecated?

📌 Answer:
 3.1 Deprecation of Legacy Authentication As of Version 9.2, NexusVortex has removed all support for alphanumeric passwords, 2-Factor Authentication (2FA) via S
MS, and standard optical fingerprint scanners. These methods were deemed "low-resonance" and vulnerable to "Ghost-Admin" takeovers.

⭐ Groundedness Score: 5/5
```

## Conclusion

Task 10 successfully builds a foundational RAG system using embeddings and similarity search. It enables dynamic information retrieval and forms the base for advanced AI systems.